

Process Upset and Near-Miss Reporting Procedure

1. Purpose

This Standard Operating Procedure (SOP) establishes a consistent process for identifying, reporting, investigating, escalating, and tracking process upsets, near-misses, and incidents within a refinery operations team. The objective is to ensure that abnormal events are reported promptly, personnel are protected, information is preserved, root causes are identified, and corrective actions are completed and verified.

2. Scope

This procedure applies to refinery employees, contractors, operators, supervisors, maintenance personnel, engineers, and other personnel who observe, experience, or respond to an abnormal process condition or safety event. It covers process deviations, equipment failures, releases, fires, injuries, unsafe conditions, near-misses, and events that could have resulted in harm. This procedure shall be used together with the refinery emergency response, incident investigation, permit-to-work, and management-of-change systems.

3. Definitions

3.1 Process Upset: An abnormal operating condition in which a process variable, equipment condition, utility, or operating parameter moves outside its approved operating range or normal control condition. An upset may be controlled without injury or release, but it can create potential for escalation.

3.2 Near-Miss: An unplanned event or unsafe condition that did not cause injury, environmental harm, equipment damage, or loss of containment, but had the potential to do so.

3.3 Incident: An unplanned event that results in, or could require formal investigation because of, injury, illness, environmental impact, fire, explosion, significant property damage, loss of containment, or other defined consequence.

3.4 Severity Level: A classification assigned according to actual and potential consequences, including effects on people, environment, assets, production, and the public.

4. Immediate Reporting Timeline

4.1 Any employee observing an imminent danger, uncontrolled release, fire, explosion, or serious injury shall immediately raise the alarm and notify the control room or emergency response organization.

4.2 Process upsets and near-misses shall be reported to the area or shift supervisor as soon as practicable and normally within 15 minutes of recognition when safe communication is available.

4.3 Events requiring formal incident reporting shall be entered into the site reporting system as soon as practicable, preferably during the same shift and no later than 24 hours unless site rules require a shorter period.

4.4 The initial notification shall contain known facts only. Personnel shall not delay emergency actions to complete paperwork.

5. Required Report Fields

Each report shall contain, where available:

1. Date and time of event.
2. Exact location, unit, equipment, and operating area.
3. Reporter name, role, and contact information.
4. Event classification: upset, near-miss, or incident.
5. Clear factual description of what occurred.
6. Process conditions, alarms, readings, and equipment status.
7. Personnel involved or exposed, without speculation.
8. Actual consequences, including injuries, releases, damage, or production impact.
9. Potential consequences if the event had progressed.
10. Immediate actions taken to stabilize the situation.
11. Isolation, shutdown, or emergency response actions.
12. Photographs, trends, logs, drawings, samples, or other evidence where appropriate.
13. Witness names and statements.
14. Notifications made and time of notification.
15. Initial severity classification and responsible supervisor.

6. Initial Assessment and Containment

6.1 The area supervisor shall confirm that personnel are safe and that the process is in a stable condition.

6.2 Operations shall follow approved operating procedures to control the upset, isolate affected equipment, reduce inventory, or initiate shutdown when required.

6.3 Evidence shall be preserved where this can be done safely. Relevant control-room trends, alarm logs, communication

records, equipment status, and physical evidence shall not be altered unnecessarily.

6.4 If the event involves an active release, fire, explosion, injury, or other emergency, the applicable emergency response plan shall take precedence.

7. Root-Cause Analysis

7.1 Form an investigation team appropriate to the severity and complexity of the event.

7.2 Establish a factual timeline using operator logs, process trends, alarms, interviews, maintenance records, procedures, and physical evidence.

7.3 Identify the immediate cause, contributing factors, and underlying causes.

7.4 Use structured methods such as the Five Whys, fault-tree analysis, barrier analysis, or another approved investigation method.

7.5 Examine human factors, equipment condition, procedures, training, supervision, process design, management systems, and organizational conditions.

7.6 Distinguish evidence-based causes from assumptions and record unresolved uncertainties.

7.7 Develop corrective and preventive actions that address underlying causes rather than only the immediate symptom.

7.8 Obtain review and approval from the designated operations, process safety, engineering, or HSE authority according to severity.

8. Corrective Action Tracking

8.1 Every approved action shall have a unique identifier, description, responsible owner, priority, target completion date, and verification method.

8.2 Actions shall be entered into the approved corrective-action tracking system.

8.3 Owners shall provide status updates until completion.

8.4 Overdue actions shall be escalated to the responsible supervisor and department manager.

8.5 Completion shall mean that the action was implemented and its effectiveness was verified, not merely that a purchase order or work request was created.

8.6 Significant actions shall be checked after implementation to confirm that the identified risk has been reduced as intended.

8.7 Lessons learned shall be communicated to affected operations and maintenance teams and incorporated into procedures, training, or engineering controls where appropriate.

9. Escalation Matrix by Severity

Level 1 – Low: Minor process deviation or near-miss with limited potential consequence and no injury, release, or significant damage. Notify the shift or area supervisor during the shift; record and track through routine corrective action.

Level 2 – Moderate: Repeated upset, significant equipment malfunction, credible potential for injury or release, minor environmental impact, or limited property damage. Notify the operations manager and relevant HSE or process-safety representative promptly; formal review is required.

Level 3 – High: Actual or potential serious injury, significant loss of containment, major equipment damage, fire, substantial environmental impact, or major production disruption. Immediately notify site leadership, emergency response as applicable, HSE, and process safety; initiate formal investigation.

Level 4 – Critical: Fatality, life-threatening event, major fire or explosion, major uncontrolled release, severe environmental impact, or event with potential for off-site consequences. Immediately activate the emergency response and site leadership notification process, secure the area, preserve evidence, and initiate the highest-level investigation and regulatory notifications required by applicable law and site procedures.

10. Management Review and Closeout

The responsible manager shall review investigation findings, corrective actions, overdue items, and recurring trends.

Repeated events shall be assessed for systemic causes and possible management-of-change or reliability improvements. An event may be formally closed only after the investigation is approved, required actions are completed or formally risk-accepted by authorized personnel, effectiveness is verified where required, and lessons learned have been communicated.

11. Records and Confidentiality

Reports, investigation records, evidence, photographs, action logs, approvals, and communications shall be retained according to refinery document-control requirements and applicable legal obligations. Information shall be shared with personnel who require it for safety, investigation, compliance, or corrective action. Reports shall remain factual, objective, and free from unsupported blame or speculation.